

- (c) The mass of the rocket in (b) is $2.9 \times 10^6 \text{ kg}$. Assume that this mass remains constant.

For this rocket, from launch to its height at a time of 20 s after launch:

- (i) calculate the gain in gravitational potential energy ΔE_p

$$\Delta E_p = \dots\dots\dots \text{ J [2]}$$

- (ii) calculate the gain in kinetic energy ΔE_k

$$\Delta E_k = \dots\dots\dots \text{ J [2]}$$

- (iii) determine the average power output of the rocket engines. Assume that resistive forces are negligible.

$$\text{power} = \dots\dots\dots \text{ W [2]}$$

[Total: 10]

